Machine learning
math
(code)

(Data)

↓

math →

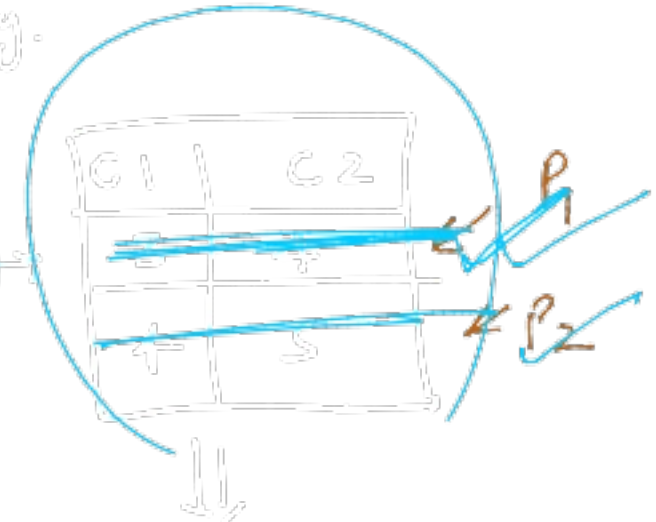
⇓

$P(a,b)$

$Q(c,d)$

$d = ?$

e.g.



OR

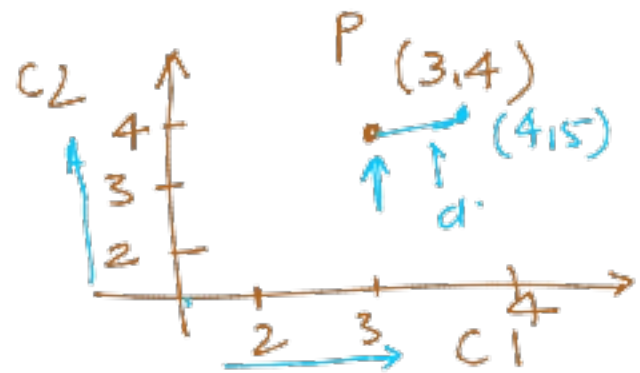
data points

⇓

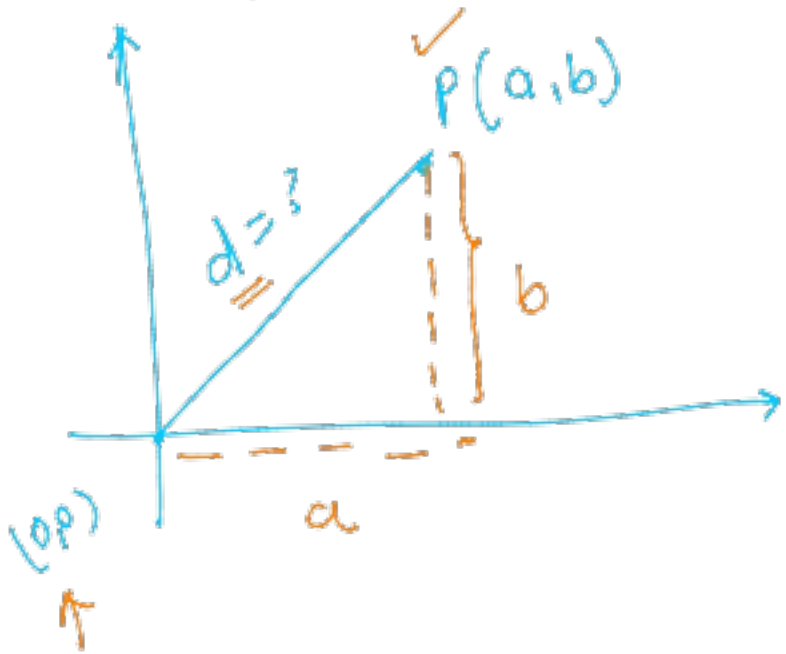
dataset

2D'

↑



distance of a points from origin →



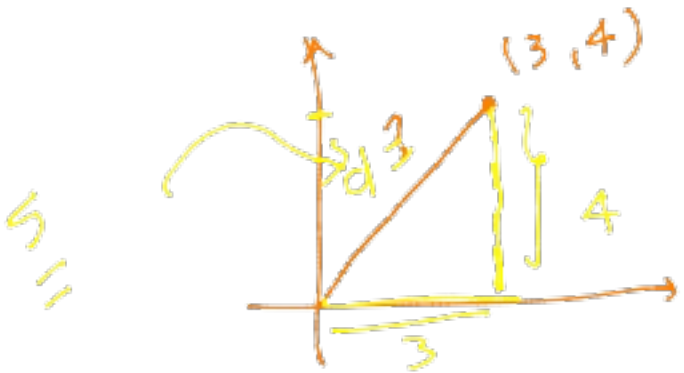
$$d^2 = a^2 + b^2$$

$$\boxed{d = \sqrt{a^2 + b^2}}$$

if your points are passing from origin

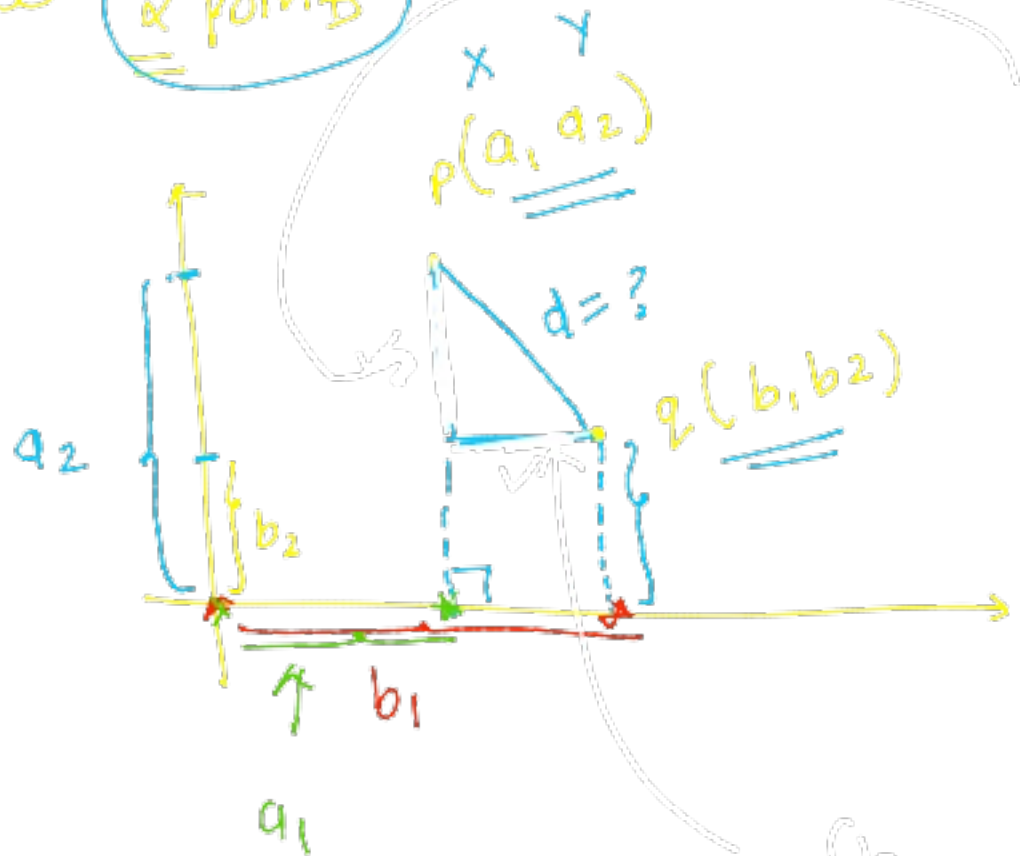
OR

$$P(a,b) \longleftrightarrow O(0,0)$$



$$\sqrt{4^2 + 3^2} \Rightarrow 5$$

distance b/w 2 points



$(a_2 - b_2)$

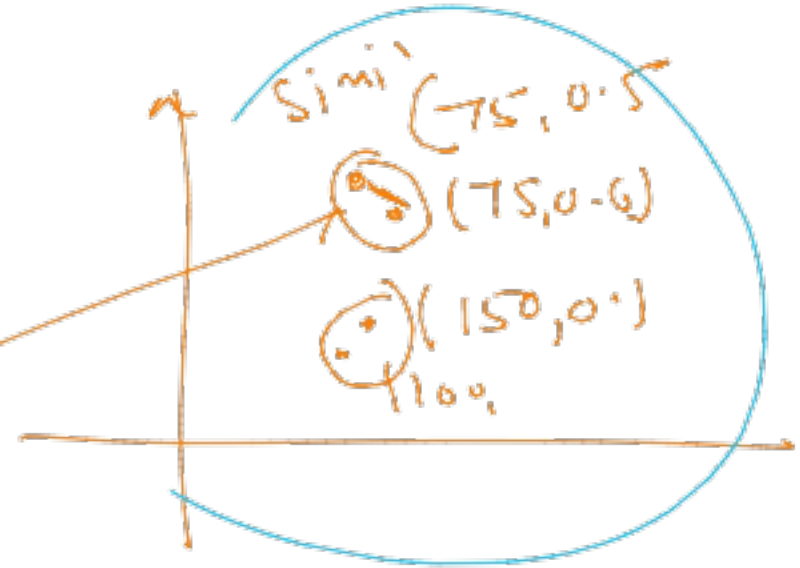
$(b_1 - a_1)$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

Euclidean distance

$$\left\{\begin{array}{l} C1 \rightarrow (75, 0.5) \\ C2 \rightarrow (75, 0.6) \\ C3 \rightarrow (150, 0.1) \\ C4 \rightarrow (160, 0.4) \end{array}\right\}$$

d(



Retail →

Euclidean formula

functions

Tx

Mumbai

USA

(lat, long)

(lat, long

Cricketing cancel time

?

[10 am]

Eu

km >>>>>> → tag

(haversine formula)

Manhattan distance

Hamming distance

DNA 1

DNA 2

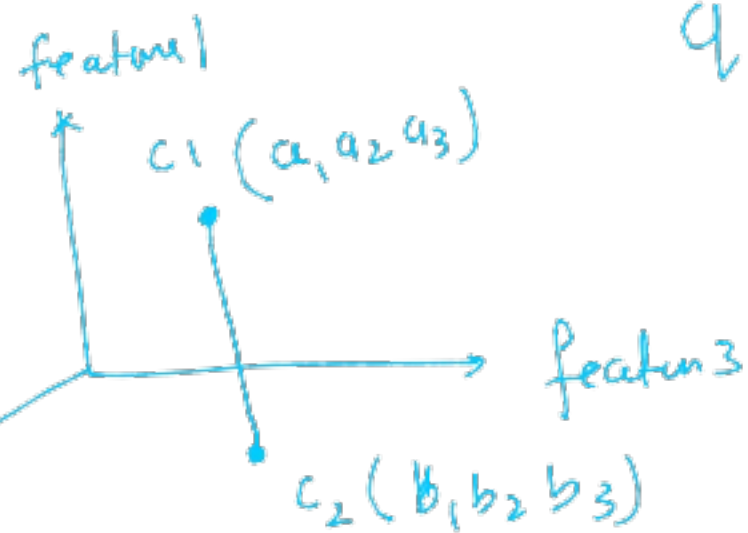
d

2D → $d = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2}$

$P(a_1, b_1)$

$q(a_2, b_2)$

(3D)



$d_i(c_1, c_2)$

$= \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2 + (a_3-b_3)^2}$

till 3D

viz ✓

Linear Alg

$d \to 0$

$$p = (a_1\ a_2\ a_3\ \cdots\ a_n)$$

$$q = (b_1\ b_2\ b_3\ \cdots\ b_n)$$

$$d = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2 + \cdots (a_n-b_n)^2}$$

for n-dimensional

$$d = \text{sqrt}\left(\sum_{i=1}^{n} (a_i - b_i)^2\right)$$

code

$\vec{a} = [a_1 \quad a_2 \quad a_3 \quad a_4 \quad - - \quad a_n]$

$\vec{b} = [b_1 \quad b_2 \quad b_3 \quad - - - \quad b_n]$

Numpy
↓
(array + arr)

(i) $a+b$ ⟹ component wise addition

eg. $p = [3, 4]$
$q = [4, 5]$
$p+q = [7, 9]$

(ii) $a*b$ → dot product
→ cross product (x) less

$a = [3, 4]$

$b = [4, 5]$

$a \cdot b = [3, 4] \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

$= 3*4 + 4*5$

$= 12 + 20$

$= 32$

Rule

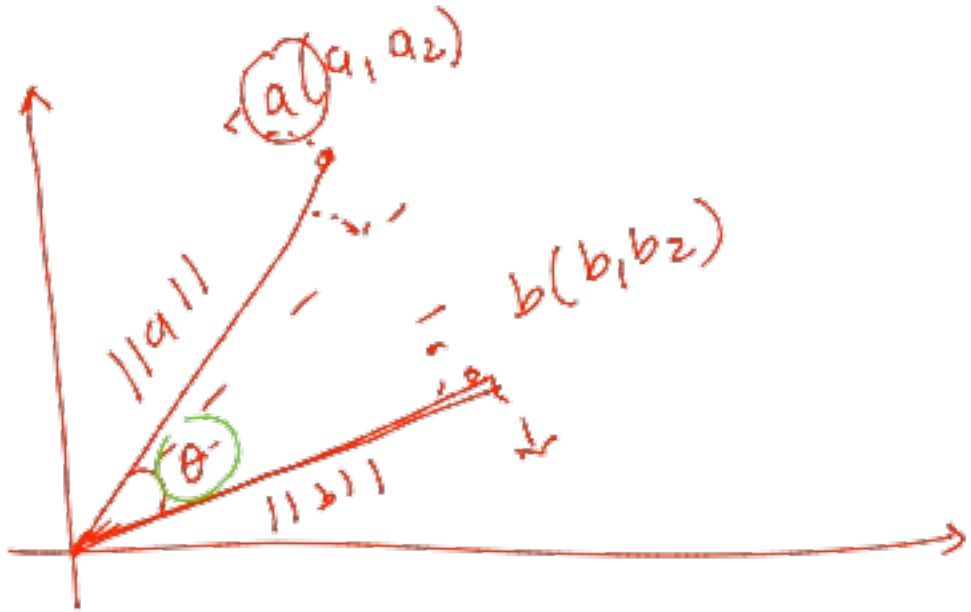
$(A)_{m \times n} \cdot (B)_{n \times p}$

$\theta \uparrow$

$\uparrow$

similarity

$\theta \downarrow$ more similar



$$a \cdot b = \underbrace{||a||}_{\text{distance of a from origin}} \, ||b|| \cdot \cos\theta$$ (formula)

$$a \cdot b = \|a\| \|b\| \cdot \cos\theta$$

$$a_1 b_1 + a_2 b_2 = \|a\| \|b\| \cdot \cos\theta$$

$$\cos\theta = \frac{a_1 b_1 + a_2 b_2}{\|a\| \|b\|}$$

cosine similarity

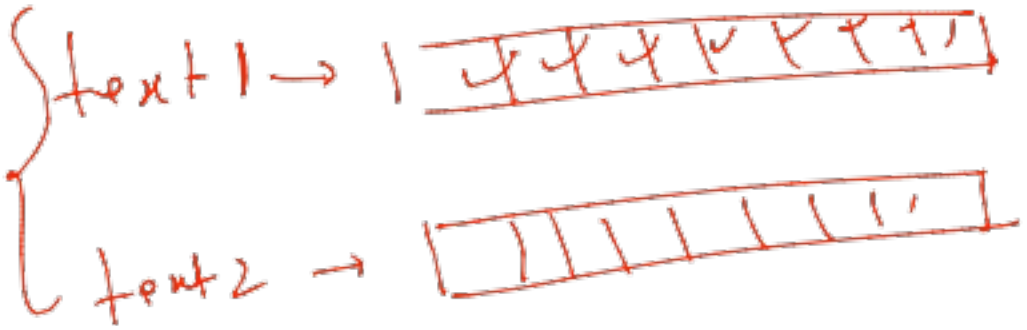
RAG

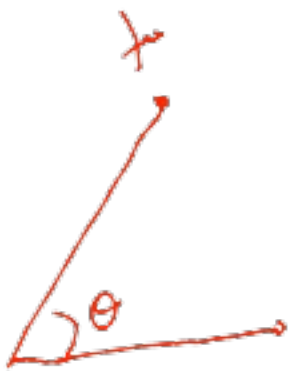
$$\boxed{\theta = \cos^{-1}\left\{\frac{a_1 b_1 + a_2 b_2}{\|a\| \|b\|}\right\}}$$

(text data)

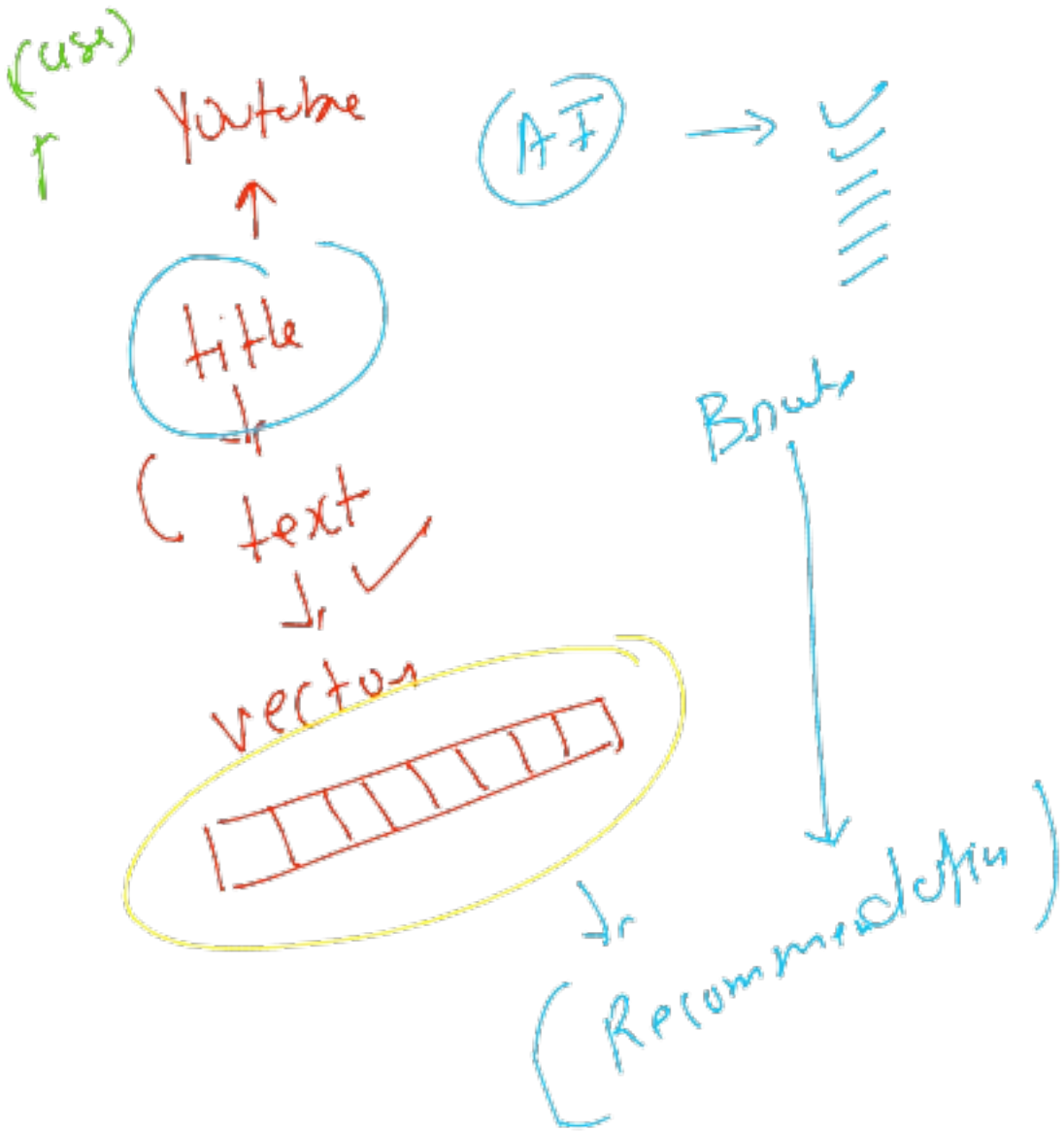
(vectors)

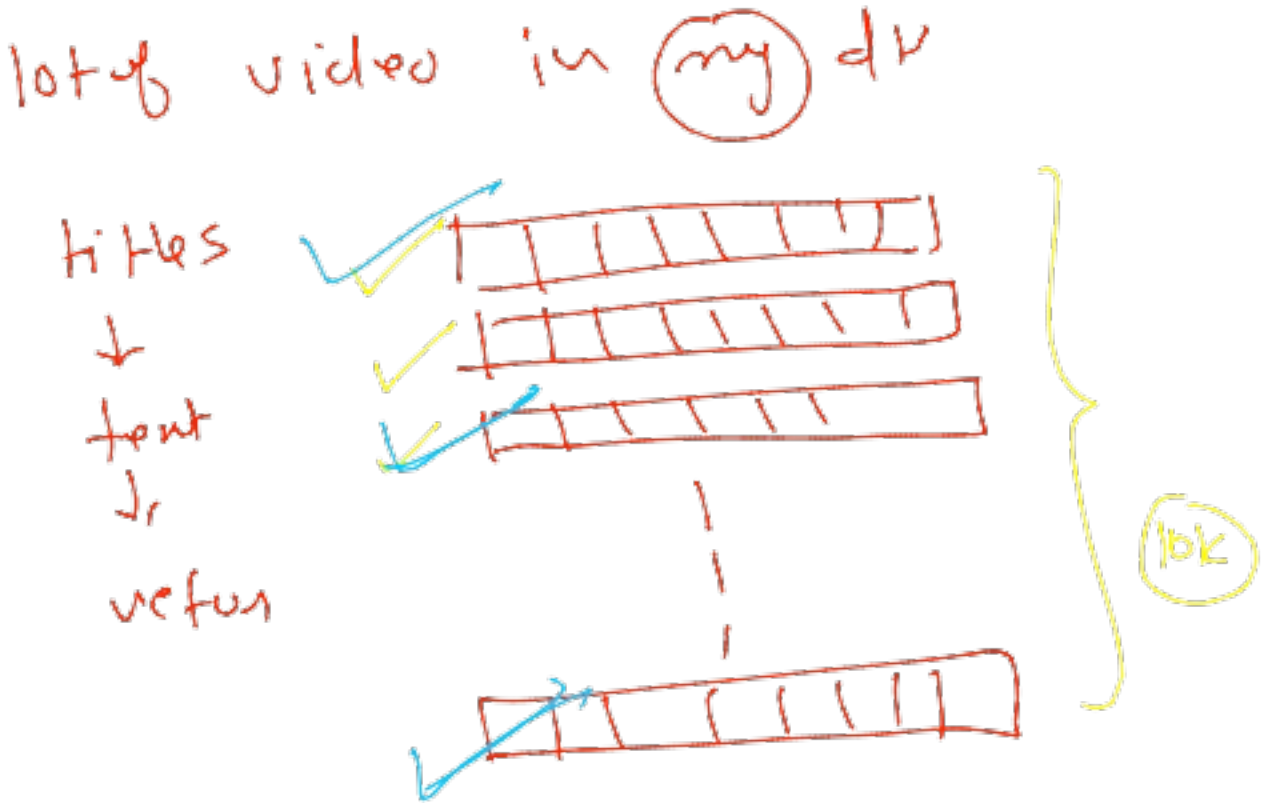
very similar





θ ↓


(user)
Youtube
AI
title
text
vector
Brute
(Recommendation)


lot of video in my dv
titles
text
vectors
10k

Line

$y = mx + c$

slope

y-intercept

m ✓

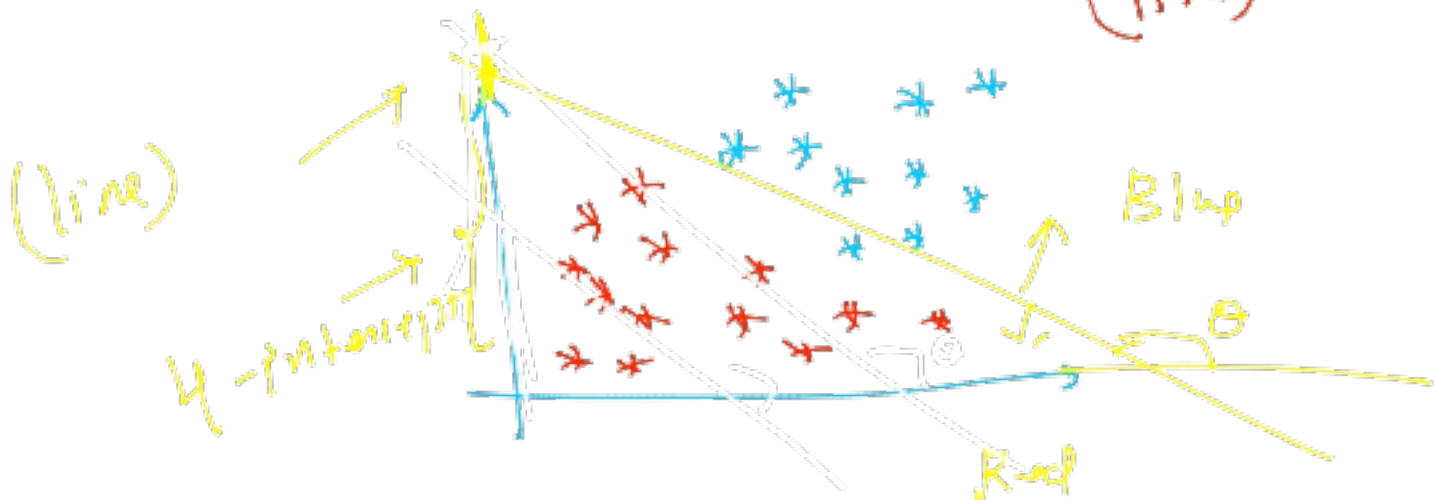
c ✓

$D_X \rightarrow$

	X	Y
c1 →	✓	✓
c2 →		
c3		
⋮	⋮	⋮
c10k		

known → log

plotted → (line)

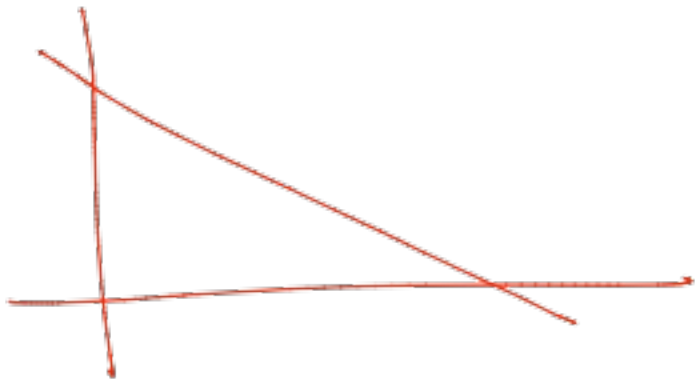


general equation of line

$$ax + by + c = 0$$

$$by = -ax - c$$

$$y = \underbrace{-\frac{a}{b}}_{m}x \underbrace{-\frac{c}{b}}_{c} \rightarrow y = mx + c$$



a b c

$$m = -\frac{a}{b}$$

$$W_1 x_1 + w_2 x_2 + w_0 = 0$$ (equation of line in 2D)

$(x_1 \;\; x_2)$ ← Data

$\boxed{(w_1 \;\; w_2) \;\; w_0}$

line parameters

$w_1 = 0.5$
$w_2 = 0.7$
$w_0 = 3$
(say)

x_1	x_2
5	4
4	5

(x_1, x_2, x_3)

(w_1, w_2, w_3) w_0

$\Downarrow$

$$\boxed{w_1 \cdot x_1 + w_2 \cdot x_2 + w_3 \cdot x_3 + w_0 = 0}$$

$\Downarrow$

nD $\rightarrow$ equal

$$w_0 + \sum_{i=1}^{n} w_i \cdot x_i = 0$$

$\Downarrow$

$$w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_0 = 0$$

dot

$\Downarrow$

nD $\rightarrow$ equal

$\Downarrow$

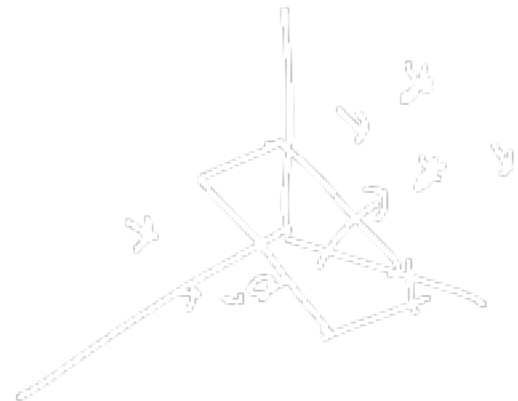
3D $\rightarrow$ line

$\Downarrow$

(plane)

$\Downarrow$

(hyperplane)



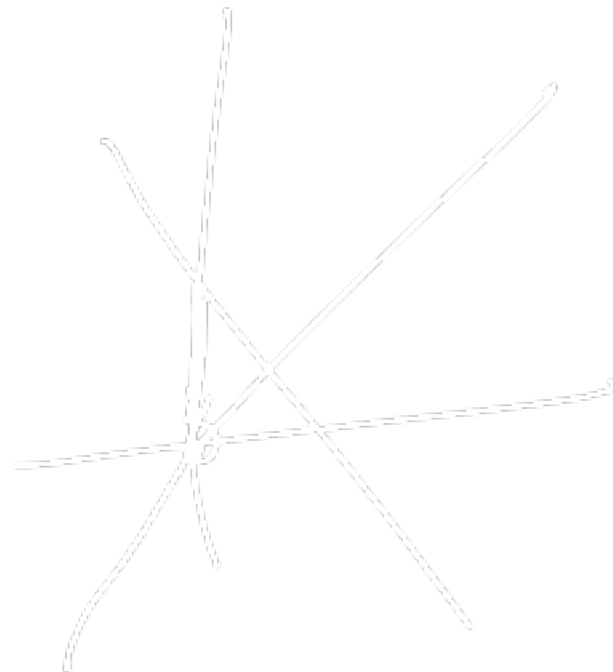
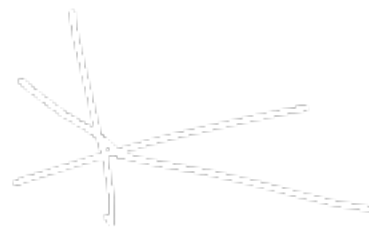
$$w_0 + \underline{[w_1\ w_2\ w_3\ \cdots\ w_n]} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

$\downarrow$

$$\boxed{\vec{w}^T \cdot x_1 + w_0 = 0}$$

$\uparrow$

$\vec{x}$

$\vec{w}$

origin